

# DMart Sales & Operational Intelligence Dashboard

## Power BI Data Analysis Project Report

### Abstract

This report presents a comprehensive analysis of DMart retail transaction data using Microsoft Power BI. The objective of the project is to transform raw transactional data into meaningful business insights through data cleaning, transformation, and visualization techniques. The project focuses on analyzing customer purchasing patterns, product performance, pricing strategies, and operational efficiency using a dataset containing more than 25,000 retail transactions.

The Power BI dashboard developed during this project highlights important key performance indicators such as total revenue, number of orders, average order value, and cancellation rate. Various visualizations such as line charts, bar charts, donut charts, and scatter plots were used to explore patterns in sales trends, customer demographics, product performance, and operational performance.

Through the use of DAX measures and Power Query transformations, the raw dataset was cleaned and transformed into a structured model suitable for business intelligence analysis. The insights obtained from this dashboard help organizations make informed business decisions and optimize retail operations.

### 1. Introduction

Retail companies generate large amounts of transactional data every day. Analyzing this data allows organizations to understand customer behavior, improve product offerings, optimize pricing strategies, and streamline operational processes.

DMart is one of the largest retail chains in India. With thousands of daily transactions across multiple locations, analyzing sales and operational data is essential to maintain competitive advantage. Business intelligence tools such as Power BI allow organizations to visualize and analyze data effectively.

In this project, a Power BI dashboard was created to analyze DMart retail data. The dashboard integrates multiple visualizations that help answer important business questions related to sales performance, customer behavior, product pricing, and operational efficiency.

The objective of this project is to demonstrate how data analytics and visualization tools can transform raw business data into actionable insights.

### 2. Project Objective

The main objective of this project is to analyze DMart retail transaction data and build an interactive Power BI dashboard that provides insights into sales performance, customer behavior, product performance, and operational efficiency.

Specific objectives include:

- Analyze customer purchasing patterns
- Identify trends in sales and revenue
- Evaluate product category performance
- Study the impact of discounts on sales
- Analyze operational metrics such as shipping mode and order status
- Build an interactive dashboard for data-driven decision making

This project aims to help stakeholders understand business performance and improve strategic decision-making.

### 3. Problem Statement

DMart wants to upgrade its decision-making process by using data-driven insights derived from historical retail transaction data.

The company aims to answer several important business questions:

#### Sales & Revenue Trends

- How do monthly and yearly sales figures compare?
- What product categories generate the highest revenue?

#### Customer Behavior

- What is the distribution of customers by age and gender?
- How do marketing interactions such as clicks affect purchase behavior?

#### Product Performance

- Which products receive the highest discounts?
- Do discounted products lead to higher sales?

#### Operational Performance

- Which shipping methods are most commonly used?
- What percentage of orders are cancelled?

#### Payment Analysis

- What is the distribution of payment methods?
- Are certain payment methods associated with higher cancellations?

The Power BI dashboard developed in this project aims to answer these questions.

### 4. Dataset Overview

The dataset used for this project contains **25,000 rows and 29 columns** representing retail transaction records.

Each row in the dataset represents a single customer transaction including customer details, product details, payment information, marketing interactions, and delivery information.

The dataset includes the following attributes:

#### Customer Demographics

- Customer Age
- Gender

#### Product Information

- Product Name
- Category
- Brand Type
- MRP
- Discount Price

#### Location Information

- State
- City
- Pin Code

#### Marketing Metrics

- Time Spent on Website
- Marketing/Advertisement Source
- Number of Clicks

#### Operational Information

- Shipping Mode
- Order Status
- Order Date
- Delivery Date
- Cancellation Date

#### Payment Information

- Payment Method
- Payment Status
- Total Order Value

These attributes allow comprehensive analysis of both customer behavior and operational performance.

### 5. Dataset Structure

The dataset contains **29 columns**, each representing specific information related to customer transactions.

The following columns are present in the dataset:

Data types of columns:

Customer ID : object

Product ID : object

Order ID : int64

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Customer Age : int64

Gender : object

Product Name : object

MRP : float64

Discount Price : float64

Category : object

State : object

City : object

Subscription : object

Bill Number : float64

Time Spent on Website : float64

Rating : float64

Marketing/Advertisement : object

Ship Mode : object

Order Status : object

Order Date : object

Delivery Date : object

Cancellation Date : object

Payment Method : object

Pin Code : int64

Total Order Value : object

Payment Status : object

No of Clicks : int64

Year : int64

Month : object

Shipping Charges : object

These columns provide a complete view of customer transactions from order placement to delivery.

## 6. Sample Data

The dataset contains records representing customer purchases across different states and cities. Each transaction includes product details, pricing information, customer demographics, and operational details.

Example fields include:

Customer ID – Unique identifier for each customer

Product ID – Unique identifier for each product

Order ID – Transaction identifier

Customer Age – Age of the customer

Gender – Male or Female

Product Name – Name of purchased product

MRP – Maximum retail price

Discount Price – Discounted product price

Category – Product category such as Branded, Local, or Imported

## 7. Tools Used

The following tools and technologies were used in this project:

Power BI Desktop

Power BI was used to build the interactive dashboard and perform data visualization.

Power Query

Power Query was used to clean and transform the raw dataset before analysis.

DAX (Data Analysis Expressions)

DAX was used to create calculated columns and measures required for analysis.

Microsoft Excel / CSV Dataset

The dataset was provided in CSV format and imported into Power BI.

## 8. Data Preparation (Power Query)

Data preparation is an important step in data analysis. Before building the dashboard, the dataset was cleaned and transformed using Power Query.

Steps performed during data preparation:

1 Import the Dataset

The dataset was imported into Power BI using

Home → Get Data → Text/CSV.

2 Open Power Query Editor

Transform Data was selected to open Power Query Editor.

3 Verify Data Types

Columns were checked to ensure correct data types such as:

Order Date → Date

Delivery Date → Date

Customer Age → Whole Number

MRP → Decimal Number

Discount Price → Decimal Number

4 Handle Missing Values

Columns were inspected for null values and cleaned accordingly.

5 Standardize Categories

Categorical columns such as gender, category, and payment method were standardized.

## 9. Data Transformation

New calculated columns were created to improve analysis.

Revenue Column

Revenue was calculated by subtracting the discount price from the original price.

Revenue = MRP – Discount Price

Delivery Days

Delivery days were calculated as the difference between order date and delivery date.

Delivery Days = Delivery Date – Order Date

These transformations helped analyze delivery efficiency and revenue trends.

## 10. DAX Measures

Several DAX measures were created in Power BI to perform calculations.

Total Revenue

SUM(Revenue)

Total Orders

COUNT(Order ID)

Average Order Value

DIVIDE([Total Revenue],[Total Orders],0)

Total Discount

SUM(Discount Price)

Cancelled Orders

CALCULATE([Total Orders], Order Status = "Cancelled")

Cancellation Rate

DIVIDE([Cancelled Orders],[Total Orders],0)\*100

Average Delivery Days

AVERAGE(Delivery Days)

These measures were used in dashboard visuals and KPI cards.

## 11. Dashboard Design

The Power BI dashboard was designed as a **single-page interactive dashboard**.

The dashboard contains the following sections:

KPI Cards

Sales Trend Charts

Customer Demographic Analysis

Product Performance Analysis

Operational Performance Metrics

Interactive slicers allow users to filter data based on different attributes such as category, gender, and payment method.

## 12. Sales Trend Analysis

Sales trends were analyzed using line charts representing monthly revenue.

This visualization helps identify:

- Seasonal sales patterns
- Peak sales periods
- Monthly performance trends

Sales trend analysis helps businesses plan marketing campaigns and inventory management strategies.

## 13. Customer Demographic Analysis

Customer demographics were analyzed using two visualizations:

Age Distribution Bar Chart

This chart shows how many customers belong to each age group.

Gender Distribution Donut Chart

This chart shows the proportion of male and female customers.

This analysis helps businesses understand their primary target audience.

## 14. Product Performance Analysis

Product performance analysis helps identify which products generate the most revenue.

Category-wise Revenue Bar Chart

Shows revenue generated by each product category.

#### Discount vs Revenue Scatter Chart

Analyzes how discount levels impact sales.

This analysis helps businesses optimize pricing strategies.

### 15. Operational Performance Analysis

Operational metrics were analyzed using:

#### Shipping Mode Usage Chart

Shows which shipping method is used most frequently.

#### Order Status Distribution

Displays number of delivered, cancelled, and pending orders.

#### Average Delivery Time

Shows average number of days taken for delivery.

These metrics help evaluate operational efficiency.

### 16. Dashboard Filters

Interactive slicers were added for:

Year

Category

Brand Type

Gender

Payment Method

These filters allow users to explore data dynamically.

### 17. Key Insights

Several important insights were obtained from the dashboard:

- Certain product categories generate higher revenue.
- Younger customers contribute significantly to sales.
- Higher discounts increase purchase probability.
- Standard shipping is the most commonly used delivery method.
- COD payments may have higher cancellation rates.

### 18. Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Optimize inventory for high-performing products
- Target marketing campaigns towards high-value customer segments
- Reduce dependency on cash-on-delivery payments
- Improve delivery speed to enhance customer satisfaction
- Implement strategic discount campaigns

## 19. Conclusion

The DMart Power BI dashboard demonstrates how business intelligence tools can transform raw data into actionable insights.

By analyzing customer behavior, product performance, and operational metrics, organizations can improve decision-making and enhance business performance.

Power BI enables businesses to visualize complex data effectively and derive valuable insights that support strategic planning and operational efficiency.

## 20. Snapshot

